SAMPLE DATA VISUALISATION QUESTIONS

Sample Theory Examination Questions

Unit I: Fundamentals of Data Visualization

1. Define data visualization and explain its importance in decision-making.
2. Explain the relationship between data analytics and data visualization.
3. Compare Tableau and Power BI.
4. Describe the steps involved in importing and preparing data in Tableau.
5. Explain the importance of collecting data from multiple sources.
6. What are the common file types used in Tableau and Power BI projects?

Concept: Data Visualization

Data visualization is the graphical representation of data using charts, graphs, maps, and dashboards to make information easier to understand and analyze.

Benefits

- Simplifies complex data
- Reveals trends and patterns
- Supports decision-making
- Improves communication of insights

Unit II: Essential Principles of Visualization Creation

7. What are the principles of effective data visualization?
8. Explain the role of data filtering in visualization.
9. What is a hierarchy in Tableau? Give an example.
10. Explain sets and their importance in data analysis.
11. Differentiate between effective and ineffective visualizations.
12. Discuss factors to consider when selecting a chart type.

Concept: Data Filtering

Filtering allows users to display only relevant data.

Example
A sales dashboard can be filtered by:

- Year
- Region
- Product category

This helps users focus on specific information.

Concept: Hierarchies

Hierarchies allow drilling down through levels of detail.

Example:

Country

↓

Province

↓

City

Users can move from a broad view to a detailed view.

Unit III: Visual Analytics with Tableau and Power BI

13. Explain the concept of visual analytics.
14. Why is decluttering important in data visualization?
15. Describe the different categories of functions used in Tableau.
16. What are aggregate functions? Give examples.
17. Explain the purpose of forecasting.
18. Discuss the role of statistics in data visualization.

Concept: Forecasting

Forecasting uses historical data to predict future values.

Example:
If sales have increased for the last 12 months, forecasting can estimate next month's sales.

Benefits:

- Planning
- Budgeting
- Resource allocation

Unit IV: R Integration with Tableau and Power BI

19. Explain the process of integrating R with Tableau.
20. What is linear regression?
21. What is Market Basket Analysis?
22. Explain the concept of K-Means Clustering.
23. Differentiate between supervised and unsupervised learning.

Concept: Linear Regression

Linear regression models the relationship between variables.

Equation:

$$Y = a + bX$$

Where:

- Y = dependent variable
- X = independent variable
- a = intercept
- b = slope

Example:
Predicting sales based on advertising expenditure.

Concept: Market Basket Analysis

Market Basket Analysis identifies products frequently purchased together.

Example:

Customer	Items

1	Bread, Milk
2	Bread, Butter
3	Bread, Milk, Butter

Rule:

Bread → Milk

Useful for:

- Product recommendations
- Store layout optimization

Concept: K-Means Clustering

K-Means groups similar data points into clusters.

Example:
A retailer may group customers into:

- High spenders
- Medium spenders
- Low spenders

Unit V: Dashboard Creation and Storytelling

24. Define a dashboard.
25. Explain the characteristics of an effective dashboard.
26. What are dashboard actions?
27. Discuss the importance of interactivity in dashboards.
28. Define data storytelling.
29. Explain how storytelling improves data communication.

Concept: Dashboard

A dashboard is a collection of visualizations that provide a summary of key information on one screen.

Characteristics:

- Interactive

- Easy to understand
- Focused on KPIs
- Real-time insights

Concept: Data Storytelling

Data storytelling combines:

1. Data
2. Visualizations
3. Narrative

to communicate insights effectively.

Unit VI: Best Practices in Data Visualization

30. Explain the art and science of data visualization.
31. Discuss principles of effective data storytelling.
32. Why should dashboards be designed for different audiences?
33. Explain limitations of Tableau and Power BI.
34. Describe common visualization mistakes.
35. Discuss current trends in data visualization.

Concept: Data Visualization Best Practices

- Keep designs simple.
- Use appropriate chart types.
- Avoid clutter.
- Maintain consistency.
- Use meaningful colors.
- Highlight key insights.

Unit VII: Real-World Data Visualization Projects

36. Discuss challenges encountered in data visualization projects.

37. Explain troubleshooting techniques in Tableau dashboards.
38. Why is feedback important in visualization projects?
39. Describe the stages of a real-world visualization project.
40. Discuss the importance of presentation skills in data analytics.

Concept: Troubleshooting

Common issues include:

- Missing data
- Incorrect filters
- Broken calculations
- Poor dashboard performance

Unit VIII: Data Visualization for Informed Decision-Making

41. Explain how dashboards support business intelligence.
42. What are parameters in Tableau and Power BI?
43. Discuss user interactivity in dashboards.
44. Explain the role of data-driven decision-making.
45. Discuss ethical issues in data visualization.
46. Explain data privacy considerations in visualization projects.

Concept: Data-Driven Decision-Making

This is the process of making decisions based on data rather than intuition.

Steps:

1. Collect data
2. Analyze data
3. Visualize results
4. Interpret insights
5. Make decisions

Concept: Data Ethics

Data ethics involves responsible handling and presentation of data.

Principles:

- Accuracy
- Transparency
- Privacy
- Fairness
- Avoid misleading visuals

Likely Long-Essay Questions (10–20 Marks)

1. Discuss the importance of data visualization in modern organizations.
2. Explain the steps involved in creating an interactive dashboard in Tableau.
3. Compare Tableau and Power BI with suitable examples.
4. Discuss the principles of effective data visualization design.
5. Explain Market Basket Analysis and its business applications.
6. Discuss data storytelling and its role in decision-making.
7. Explain ethical considerations in data visualization.
8. Describe how forecasting can be used to support business decisions.
9. Discuss dashboard interactivity and parameterization in Tableau and Power BI.
10. Explain K-Means Clustering and its applications in visual analytics.

46 exam questions based on the DS231 course outline.

Unit I: Fundamentals of Data Visualization

1. Define data visualization and explain its importance.

Data visualization is the graphical representation of data using charts, graphs, maps, and dashboards. It helps users identify trends, patterns, and insights quickly, supporting informed decision-making.

2. Relationship between data analytics and data visualization.

Data analytics involves examining data to discover insights, while data visualization presents those insights visually for easier understanding and communication.

3. Compare Tableau and Power BI.

Tableau	Power BI
Strong visualization capabilities	Strong Microsoft integration
Easy drag-and-drop interface	Cost-effective
Widely used for analytics	Widely used for business intelligence

4. Steps involved in importing and preparing data.

1. Connect to data source.
2. Import data.
3. Clean data.
4. Transform data.
5. Validate data.
6. Create visualizations.

5. Importance of collecting data from multiple sources.

It provides comprehensive analysis, improves accuracy, and allows richer insights.

6. Common file types.

Excel files, CSV files, databases, JSON files, text files, and cloud data sources.

Unit II: Essential Principles of Visualization Creation

7. Principles of effective data visualization.

- Simplicity
- Accuracy
- Clarity
- Consistency
- Relevance
- Proper chart selection

8. Role of data filtering.

Filtering allows users to focus on specific data subsets and remove irrelevant information.

9. What is a hierarchy?

A hierarchy organizes data into levels for drill-down analysis.

Example:
Country → Province → District → City

10. What are sets?

Sets are custom groups of data points used to compare categories or highlight specific records.

11. Effective vs ineffective visualizations.

Effective visualizations:

- Clear
- Accurate
- Easy to understand

Ineffective visualizations:

- Cluttered
- Misleading
- Difficult to interpret

12. Factors when selecting a chart.

- Type of data
- Analysis objective
- Audience
- Number of variables
- Desired comparison

Unit III: Visual Analytics

13. Explain visual analytics.

Visual analytics combines analytical techniques with interactive visualizations to gain insights from data.

14. Why is decluttering important?

It improves readability and helps users focus on important information.

15. Categories of functions.

- Number functions
- String functions
- Logical functions
- Date functions
- Aggregate functions

16. What are aggregate functions?

Functions that summarize data.

Examples:

- SUM()
- AVG()
- COUNT()
- MIN()
- MAX()

17. Purpose of forecasting.

Forecasting predicts future outcomes based on historical data.

18. Role of statistics in visualization.

Statistics help summarize, analyze, and interpret data accurately.

Unit IV: R Integration

19. Process of integrating R with Tableau and Power BI.

1. Install R.
2. Configure Tableau or Power BI to locate R.
3. Create R scripts.
4. Use R outputs in visualizations.

20. What is linear regression?

A statistical technique used to model relationships between dependent and independent variables.

21. What is Market Basket Analysis?

A technique used to identify items frequently purchased together.

22. Explain K-Means Clustering.

An unsupervised learning method that groups similar data points into clusters.

23. Supervised vs Unsupervised Learning.

Supervised	**Unsupervised**
Uses labeled data	Uses unlabeled data
Prediction focused	Pattern discovery focused
Example: Regression	Example: K-Means

Unit V: Dashboard Creation and Storytelling

24. Define a dashboard.

A dashboard is a collection of visualizations displayed on one screen to monitor and analyze key information.

25. Characteristics of an effective dashboard.

- Interactive
- Clear
- Accurate
- Relevant
- Easy to navigate

26. What are dashboard actions?

Interactive features that allow users to filter, highlight, or navigate between views.

27. Importance of interactivity.

It improves user engagement and allows deeper exploration of data.

28. Define data storytelling.

The process of combining data, visuals, and narrative to communicate insights effectively.

29. How storytelling improves communication.

It makes complex information understandable, memorable, and actionable.

Unit VI: Best Practices

30. Art and science of data visualization.

Art focuses on design and aesthetics, while science focuses on accurate representation of data.

31. Principles of effective data storytelling.

- Know the audience
- Focus on key message
- Use appropriate visuals
- Maintain logical flow

32. Why design for different audiences?

Different audiences have different information needs and levels of expertise.

33. Limitations of Tableau and Power BI.

- Performance issues with large datasets
- Licensing costs
- Learning curve
- Dependency on data quality

34. Common visualization mistakes.

- Too many colors
- Cluttered dashboards
- Wrong chart selection
- Missing labels

35. Current trends in visualization.

- Interactive dashboards
- AI-assisted analytics

- Real-time reporting
- Mobile visualizations

Unit VII: Real-World Projects

36. Challenges in visualization projects.

- Poor data quality
- Missing data
- Complex datasets
- Time constraints

37. Troubleshooting techniques.

- Check data connections
- Validate calculations
- Verify filters
- Review dashboard settings

38. Why is feedback important?

Feedback helps improve visualization quality and usability.

39. Stages of a visualization project.

1. Data collection
2. Data preparation
3. Analysis
4. Visualization
5. Presentation
6. Evaluation

40. Importance of presentation skills.

They help communicate insights effectively to stakeholders.

Unit VIII: Decision-Making

41. How dashboards support business intelligence.

Dashboards provide real-time insights that help organizations monitor performance and make informed decisions.

42. What are parameters?

Parameters are user-controlled variables that allow dynamic changes in visualizations.

43. User interactivity in dashboards.

Users can filter, drill down, highlight, and modify views to explore data.

44. Role of data-driven decision-making.

It enables decisions based on evidence and analysis rather than assumptions.

45. Ethical issues in data visualization.

- Misleading charts
- Data manipulation
- Privacy violations
- Biased presentation

46. Data privacy considerations.

Organizations must protect sensitive information, obtain consent where necessary, and comply with privacy regulations.

High-Probability Exam Definitions

- **Dashboard:** A collection of visualizations on a single screen.
- **Filter:** A tool that restricts displayed data.
- **Hierarchy:** A structure that supports drill-down analysis.
- **Set:** A custom grouping of data.
- **Forecasting:** Predicting future values using historical data.
- **Linear Regression:** Modeling the relationship between variables.
- **Market Basket Analysis:** Finding products frequently bought together.
- **K-Means Clustering:** Grouping similar records into clusters.
- **Parameter:** A user-controlled variable.
- **Data Storytelling:** Communicating insights through data, visuals, and narrative.

- **Business Intelligence:** Using data to support organizational decisions.

10 likely long-essay questions 10–20 marks.

1. Discuss the Importance of Data Visualization in Modern Organizations

Data visualization is the graphical representation of data through charts, graphs, maps, dashboards, and reports. It plays a critical role in modern organizations because it transforms complex data into understandable information.

Firstly, data visualization helps decision-makers quickly identify trends, patterns, and anomalies. Instead of analyzing large tables of data, managers can use dashboards to understand business performance at a glance.

Secondly, visualization improves communication among stakeholders. Visual reports make it easier to explain findings to both technical and non-technical audiences.

Thirdly, organizations use visualizations to monitor Key Performance Indicators (KPIs), track progress, and evaluate business objectives. Real-time dashboards allow organizations to respond quickly to changes.

Finally, data visualization supports data-driven decision-making by providing evidence-based insights rather than relying on assumptions.

Conclusion: Data visualization improves understanding, communication, and decision-making, making it an essential tool for modern organizations.

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2. Explain the Steps Involved in Creating an Interactive Dashboard in Tableau

An interactive dashboard combines multiple visualizations into a single interface that allows users to explore data.

Steps:

1. Connect Tableau to the data source.
2. Import and clean the data.
3. Create worksheets such as bar charts, line charts, maps, and pie charts.
4. Build calculated fields if needed.
5. Create filters and parameters.
6. Create a dashboard and add worksheets.

7. Arrange visualizations logically.
8. Add dashboard actions such as filter actions, highlight actions, and URL actions.
9. Format the dashboard for clarity and consistency.
10. Test interactivity and publish the dashboard.

Benefits:

- Improved user engagement.
- Faster decision-making.
- Better data exploration.

Conclusion: Interactive dashboards provide a powerful way to monitor and analyze business performance.

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3. Compare Tableau and Power BI with Suitable Examples

Tableau and Power BI are popular data visualization tools used for business intelligence.

Feature	Tableau	Power BI
Developer	Tableau Software	Microsoft
Visualization	Very advanced	Strong but slightly less flexible
Cost	More expensive	More affordable
Integration	Multiple platforms	Excellent Microsoft integration
Ease of Use	User-friendly	Familiar to Excel users

Similarities

- Both create dashboards and reports.
- Both support data connections from multiple sources.
- Both provide interactive visualizations.

Example

A retail company can use Tableau or Power BI to visualize monthly sales, customer behavior, and inventory performance.

Conclusion: Tableau excels in visualization flexibility, while Power BI offers strong integration with Microsoft products and lower costs.

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4. Discuss the Principles of Effective Data Visualization Design

Effective data visualization ensures that information is presented clearly and accurately.

Key Principles

1. **Simplicity** – Avoid unnecessary elements.
2. **Clarity** – Ensure charts are easy to understand.
3. **Accuracy** – Represent data truthfully.
4. **Consistency** – Use consistent colors, fonts, and layouts.
5. **Appropriate Chart Selection** – Choose charts suited to the data.
6. **Visual Hierarchy** – Highlight important information.
7. **Accessibility** – Ensure readability for all users.

Importance

These principles help users interpret data quickly and reduce the risk of misunderstanding.

Conclusion: Good design improves communication and helps audiences derive meaningful insights from data.

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5. Explain Market Basket Analysis and Its Business Applications

Market Basket Analysis is a data mining technique used to identify products that are frequently purchased together.

How It Works

The technique analyzes transaction data to discover associations between products.

Example:

Transaction	Items
T1	Bread, Milk
T2	Bread, Butter
T3	Bread, Milk, Butter

Possible Rule:
Bread → Milk

Business Applications

- Product recommendations.
- Store layout optimization.
- Cross-selling strategies.
- Promotional campaigns.
- Inventory planning.

Benefits

- Increased sales.
- Improved customer experience.
- Better marketing decisions.

Conclusion: Market Basket Analysis helps organizations understand customer purchasing patterns and improve profitability.

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6. Discuss Data Storytelling and Its Role in Decision-Making

Data storytelling is the practice of combining data, visualizations, and narrative to communicate insights effectively.

Components

1. Data
2. Visualizations
3. Story or narrative

Importance

- Makes complex information easier to understand.
- Engages audiences.
- Supports informed decisions.
- Improves communication of analytical findings.

Example

A sales dashboard may show declining sales. A story can explain when the decline started, possible causes, and recommended actions.

Conclusion: Data storytelling transforms raw data into meaningful information that supports decision-making.

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7. Explain Ethical Considerations in Data Visualization

Ethics in data visualization involves presenting data honestly and responsibly.

Ethical Principles

1. Accuracy.
2. Transparency.
3. Fairness.
4. Privacy.
5. Accountability.

Common Ethical Issues

- Manipulating scales to exaggerate trends.
- Omitting important information.
- Using misleading chart types.
- Revealing confidential information.

Importance

Ethical visualization maintains trust and ensures decisions are based on truthful information.

Conclusion: Analysts must present data fairly and protect privacy to maintain credibility and integrity.

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8. Describe How Forecasting Can Be Used to Support Business Decisions

Forecasting uses historical data to predict future outcomes.

Process

1. Collect historical data.
2. Identify trends and patterns.
3. Apply forecasting techniques.
4. Generate predictions.

5. Use predictions for planning.

Applications

- Sales forecasting.
- Demand planning.
- Budget preparation.
- Resource allocation.
- Inventory management.

Benefits

- Reduces uncertainty.
- Improves planning.
- Supports strategic decisions.

Conclusion: Forecasting enables organizations to prepare for future opportunities and challenges.

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9. Discuss Dashboard Interactivity and Parameterization in Tableau and Power BI

Dashboard interactivity allows users to explore data dynamically.

Interactive Features

- Filters
- Drill-downs
- Highlighting
- Tooltips
- Dashboard actions

Parameters

Parameters are user-defined variables that allow dynamic control over visualizations.

Benefits

- Personalized analysis.
- Better exploration of data.

- Improved user engagement.
- Faster insight discovery.

Example

A parameter may allow users to switch between viewing Sales, Profit, and Quantity on the same dashboard.

Conclusion: Interactive dashboards improve analytical capabilities and user experience.

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10. Explain K-Means Clustering and Its Applications in Visual Analytics

K-Means Clustering is an unsupervised machine learning technique that groups similar data points into clusters.

Steps

1. Choose the number of clusters (K).
2. Select initial centroids.
3. Assign data points to the nearest centroid.
4. Recalculate centroids.
5. Repeat until clusters stabilize.

Applications

- Customer segmentation.
- Market research.
- Fraud detection.
- Recommendation systems.
- Pattern recognition.

Advantages

- Easy to implement.
- Efficient for large datasets.
- Helps discover hidden patterns.

Limitations

- Requires choosing K beforehand.

- Sensitive to outliers.

Conclusion: K-Means Clustering is a valuable technique for discovering groups and patterns within data.

Exam Tip

For a 15–20 mark question, structure your answer as:

1. Introduction/Definition
2. Main Explanation (3–6 points)
3. Examples/Applications
4. Advantages or Importance
5. Conclusion

This structure usually earns higher marks because it shows both understanding and organization.

Question : What are the principals of data filtering?

Exam Definition

Data filtering is the process of selecting specific subsets of data based on defined conditions so that users can focus on relevant information and gain clearer insights from visualizations.

1. Relevance

Only display data that is relevant to the user's objective or analysis.

Example: A sales manager may filter sales data by region to focus on their area of responsibility.

2. Simplicity

Filters should be easy to understand and use. Avoid too many unnecessary filter options.

3. Accuracy

Filtering should not distort or misrepresent the data. The filtered results must accurately reflect the selected criteria.

4. Consistency

Similar filters should behave the same way across all charts and dashboards.

Example: If selecting "2025" filters one chart, it should filter related charts consistently.

5. Flexibility

Users should be able to filter data using different criteria such as:

- Date
- Region
- Category
- Product
- Customer

6. Performance

Filters should return results quickly, especially when working with large datasets.

7. User Control

Users should be able to apply, modify, or remove filters easily to explore different perspectives of the data.

Types of Filters in Tableau and Power BI

- **Dimension Filters** – Filter categorical data (e.g., Region, Gender).
- **Measure Filters** – Filter numerical values (e.g., Sales > 1000).
- **Date Filters** – Filter by year, month, or day.
- **Context Filters (Tableau)** – Create a primary filter that other filters depend on.
- **Visual/Page/Report Filters (Power BI)** – Apply filtering at different levels.